

COMP6203 - INTELLIGENT AGENTS

Introduction to the Module

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Introduction to COMP6203

WELCOME

- Welcome to COMP6203 Intelligent Agents!
- This lecture is being recorded and will be available for you to rewatch at the end of the session
- Links to all the information regarding this module, including slides, videos etc. can be found at the following link:

<https://moodle.ecs.soton.ac.uk/course/view.php?id=850>

- If you are auditing this module, we've given you authorisation to do so, and you do not have access to the material, please email me for access

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WHAT ARE INTELLIGENT AGENTS

- The field of Intelligent Agents is about *autonomous decision making* based on available information
- Intelligent Agents make decisions on behalf of people, groups or organisations
- Intelligent Agents interact with other agents in a multi-agent system, but also humans
- Intelligent Agents have specific goals such as completing tasks, and often compete for scarce resources with other agents

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RELATION TO OTHER AI FIELDS

- Intelligent Agents vs Robotics.

Whereas robotics focuses on hardware and physical presence, intelligent agents are about autonomous “software”

- Intelligent Agents vs Machine Learning.

Whereas machine learning is about deriving useful information from data, intelligent agents is making (optimal) decisions based on this information.

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WHY STUDY INTELLIGENT AGENTS?

- Finance/e-commerce
- The Smart Grid
- Autonomous Vehicles
- Robotics
- Kidney Exchange
- Security Patrolling
- Matching
- Adversarial Machine Learning
- ...

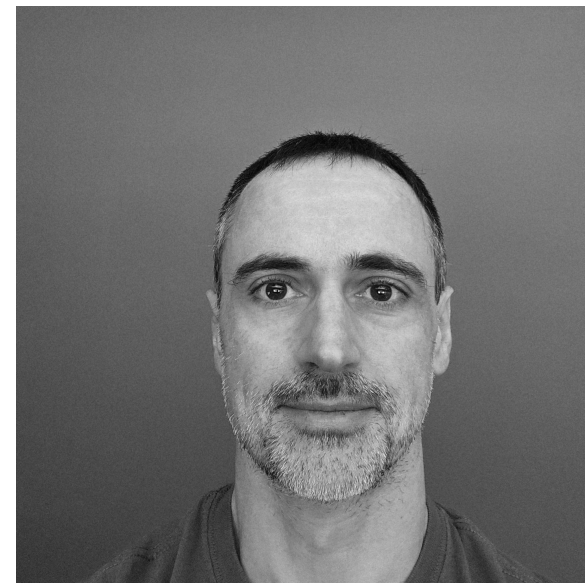
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MAIN TOPICS COVERED

- Modelling agent preferences: *utility theory*
- Multi-agent strategic interactions: *game theory*
- Conflict resolution and resource allocation: *negotiation, voting, and auctions*
- Multi-Agent Collaboration: *coalition formation*

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LECTURERS, ROLES AND TOPICS



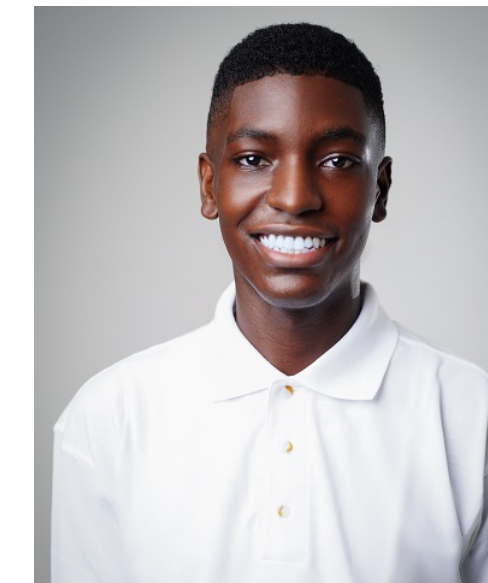
Dr Enrico Marchioni
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(Module Leader)

- Utility theory
- Game theory
- Coalition formation



Dr Bahar Rastegari
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- Auctions
- Voting
- Negotiation



Keniel Peart
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- Coursework assignment

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MODULE DELIVERY OVERVIEW

- All of the technical contents is delivered through pre-recorded lectures (with the exception of tomorrow's lecture and other lectures to be announced)
- Videos will be uploaded on a weekly basis: they will be available through Panopto (links on module page)
- In addition, there are weekly in-person tutorial sessions:
 - **Mondays 09:00 (46/2005)**
 - **Fridays 10:00 (100/4011)**
 - Aim is to discuss pre-recorded lectures with Q&A, exercises etc.
 - Some weeks will have only one tutorial session (check module page!)
 - Some tutorial sessions will be used to provide feedback for the tests
- Labs **Tuesdays 14:00-16:00** (59/3229 ECS Computing Lab)
 - These are in-person sessions
 - No labs in Week 1 and Week 2
 - They are for assessments and for LAB sessions for the coursework (more about this later)

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TUTORIAL SESSIONS

- Aim: discuss the contents of the weekly pre-recorded theoretical lectures through Q&A
- Each week a worksheet with exercises will be available: you should attempt to solve the exercises
- Tutorial sessions will be used to go through/solve exercises
- It is fundamental that you watch the pre-recorded lectures **BEFORE** attending the tutorial
- If you have any questions about the lectures, you should get in touch before the tutorial so that they can be properly addressed during the session

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ASSESSMENT

- There are 3 types of assessments:
- 3 Tests - worth **10%** in total
 - Designed to make sure you keep up to speed with pre-recorded lectures
 - Each assessment will cover a different subset of topics
 - You will receive feedback on the assessments, and topics that proved challenging will be discussed during one of the tutorial sessions
 - **Weeks 6, 9 and 11**
- Coursework Assignment – worth **40%**
- Final Assessment (Exam) – worth **50%**

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COURSEWORK ASSIGNMENT

- Develop a trading agent for a maritime competition
- Agent needs to be written in the Python
 - We will not introduce Python as part of the module
 - Start now with e.g. sessions from LinkedIn learning (see module page for details)
- Group work:
 - 4 students per group
 - Find groups now and fill in group survey (TBA)
 - Submit group details by **Friday October 9, 14:00**
- Assessment:
 - Performance of the agent in a class competition
 - Group report displaying understanding of the task, design, analysis and implementation of the agent
 - Details will be discussed next week

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GETTING HELP / EXPECTATIONS

- Coursework
 - Tuesday's sessions
- Lectures
 - Tutorials
 - E-mail
- Setting Expectations:
 - Don't expect response out of office hours, during weekends and holidays
 - Might take up to 2 working days to get email response in busy times
 - Send reminder if you see away message or have not received response after more than 2 working days

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RESOURCES

- Slides+Website
- **An Introduction to MultiAgent Systems.** M. J. Wooldridge. John Wiley & Sons; 2nd Edition edition, 2009.
- **Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations.** Y. Shoham, K. Leyton-Brown. Cambridge University Press, 2009.
<http://www.masfoundations.org/mas.pdf>
- **Game Theory** (2nd ed.). M. Maschler, E. Solan, S. Zamir. Cambridge University Press, 2020.
- **Algorithmic Game Theory.** N. Nisan, T. Roughgarden, E. Tardos, V. Vazirani. Cambridge University Press, 2007.
http://www.cambridge.org/journals/nisan/downloads/Nisan_Non-printable.pdf

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Q&A

QUESTIONS?